

Week 4: Data Preparation and My First Web App

Key Concepts:

- **Feature Engineering:** Constructing new financial metrics from raw data, specifically calculating the **Altman Z-Score** (vectorized approach) and **Average Cost of Debt**.
- **Time-Series Logic in Panel Data:** Understanding "lagged" variables (e.g., using last year's credit risk to predict this year's borrowing cost) using sorting and shifting operations.
- **Exploratory Data Analysis (EDA):** Using contingency tables (crosstabs) and statistical regression (`statsmodels`) to validate relationships between risk and cost.
- **Reactivity:** The core concept of marimo where UI inputs (sliders, dropdowns) immediately trigger code execution and update visualizations.
- **Web Assembly (WASM) & Deployment:** The ability to convert a Python notebook into a standalone HTML file that runs entirely in the browser (client-side) and hosting it via GitHub Pages.

Coding Techniques:

- **Vectorization:** Writing functions (like `Zscore`) that operate on entire Pandas Series at once rather than loops.
- **Chaining Methods:** Combining `sort_values()` , `groupby()` , and `shift()` in a single line to calculate year-over-year changes per company.
- **Reactive Data Filtering:** Creating a dataset variable that listens to a UI widget and updates automatically: `df_filtered = df[df['Sector'].isin(selector.value)]` .
- **Layout Construction:** Assembling a multi-tabbed web page by assigning markdown and plot objects to variables and passing them into a `mo.ui.tabs` dictionary.

Packages Introduced:

- **Pandas:** extensively used for `read_csv` , `fillna` , `dropna` , `groupby` , `shift` , `sort_values` , and `crosstab` .
- **Plotly Express:** Used for advanced statistical plots (Box plots with outliers) and interactive maps (`scatter_geo`).
- **Marimo:** Used for creating UI widgets (`mo.ui.slider` , `mo.ui.multiselect` , `mo.ui.tabs`) and layout management (`mo.vstack` , `mo.hstack` , `mo.md`).
- **Statsmodels:** Used for running an Ordinary Least Squares (OLS) regression.